

DIFFERENTIATION

- **Definition:** Let $y = f(x)$ be a real function defined in $(x-d, x+d)$ which is a neighborhood of real number x .
 - (1) $\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ if it exists, is called as the derivative of $f(x)$ and is denoted by $f'(x)$ or $\frac{dy}{dx}$.
 - (2) $\lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h}$ if it exists, is called as the right hand derivative of $f(x)$ and is denoted by $Rf'(x)$.
 - (3) $\lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h}$ if it exists, is called as the left hand derivative of $f(x)$ and is denoted by $Lf'(x)$.
 - (4) If $Lf'(x)$ and $Rf'(x)$ both exist and $Lf'(x) = Rf'(x)$, then $f(x)$ is differentiable at x .
- **Differentiability on an interval:** A function $f(x)$, defined on $[a, b]$ is said to be differentiable on $[a, b]$, if it is differentiable at every $c \in [a, b]$ and both $\lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}$ and $\lim_{h \rightarrow 0^-} \frac{f(b-h) - f(b)}{-h}$ exists.
- **Standard results:**
 - (a) Every polynomial function is differentiable at each $x \in R$.
 - (b) Every exponential function a^x , $a > 0$ is differentiable at each $x \in R$.
 - (c) Every constant function is differentiable at each $x \in R$.
 - (d) Logarithmic functions are differentiable in their domains.
 - (e) Trigonometric functions are differentiable in their domains.
 - (f) Inverse Trigonometric functions are differentiable in their domains.
 - (g) The sum of two differentiable functions is differentiable.
 - (h) The difference of two differentiable functions is differentiable.
 - (i) The product of two differentiable functions is differentiable.
 - (j) The quotient of two differentiable functions is differentiable.
 - (k) The composition of differentiable functions (under conditions) is a differentiable.

• **Derivative of Standard Functions (All Formulae)**

1. $\frac{d}{dx}(k) = 0$, where k is a constant.
2. $\frac{d}{dx}(x^n) = nx^{n-1}$, where $n \in R$
3. $\frac{d}{dx}((ax+b)^n) = na(ax+b)^{n-1}$, where $n \in R$
4. $\frac{d}{dx}(e^x) = e^x$
5. $\frac{d}{dx}(a^x) = a^x \ln a$
6. $\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$
7. $\frac{d}{dx}(\ln x) = \frac{d}{dx}(\log_e x) = \frac{1}{x}$
8. $\frac{d}{dx}(|x|) = \frac{x}{|x|}$, $x \neq 0$ as $|x|$ is not differentiable at $x = 0$.
9. $\frac{d}{dx}([x]) = \begin{cases} 0 & x \in R - Z \\ \text{does not exist} & x \in Z \end{cases}$.
10. $\frac{d}{dx}(\{x\}) = \begin{cases} 1 & x \in R - Z \\ \text{does not exist} & x \in Z \end{cases}$.
11. $\frac{d}{dx}(\sin x) = \cos x$
12. $\frac{d}{dx}(\cos x) = -\sin x$
13. $\frac{d}{dx}(\tan x) = \sec^2 x$
14. $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$
15. $\frac{d}{dx}(\sec x) = \sec x \cdot \tan x$
16. $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cdot \cot x$
17. $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
18. $\frac{d}{dx}(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$
19. $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

20. $\frac{d}{dx}(\cot^{-1} x) = \frac{-1}{1+x^2}$
21. $\frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{-1}{x\sqrt{x^2-1}}$
22. $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$
23. $\frac{d}{dx}(\sinh x) = \cosh x$
24. $\frac{d}{dx}(\cosh x) = \sinh x$
25. $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$
26. $\frac{d}{dx}(\coth x) = -\operatorname{cosech}^2 x$
27. $\frac{d}{dx}(\operatorname{sech} x) = -\operatorname{sech} x \cdot \tanh x$
28. $\frac{d}{dx}(\operatorname{cosech} x) = -\operatorname{cosech} x \cdot \coth x$
29. $\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$
30. $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$
31. $\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$
32. $\frac{d}{dx}(\coth^{-1} x) = \frac{1}{1-x^2}$
33. $\frac{d}{dx}(\operatorname{cosech}^{-1} x) = \frac{-1}{x\sqrt{x^2+1}}$
34. $\frac{d}{dx}(\operatorname{sech}^{-1} x) = \frac{-1}{x\sqrt{1-x^2}}$
35. $\frac{d}{dx}(a \cdot f(x)) = a \cdot \frac{d}{dx}(f(x))$, where a is a constant.
36. $\frac{d}{dx}(f(x) + g(x)) = \frac{d}{dx}(f(x)) + \frac{d}{dx}(g(x))$
37. $\frac{d}{dx}(f(x) - g(x)) = \frac{d}{dx}(f(x)) - \frac{d}{dx}(g(x))$
38. $\frac{d}{dx}(f(x) \times g(x)) = f(x) \times \frac{d}{dx}(g(x)) + g(x) \times \frac{d}{dx}(f(x))$

39. $\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x) \times \frac{d}{dx}(f(x)) - f(x) \times \frac{d}{dx}(g(x))}{\{g(x)\}^2}$
40. $\frac{d}{dx}(f \pm g \pm h \pm \dots) = \frac{d}{dx}(f) \pm \frac{d}{dx}(g) \pm \frac{d}{dx}(h) \pm \dots$
41. If $y = f(t)$, $t = g(x)$, then $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$
42. $\frac{dy}{dx} \times \frac{dx}{dy} = 1$.
43. $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$.
44. If $y = (f(x))^{g(x)}$, then $\frac{dy}{dx} = (f(x))^{g(x)} \left[g(x) \frac{f'(x)}{f(x)} + g'(x) \log f(x) \right]$.
45. If $y = f(t)$, $x = g(t)$, then $\frac{dy}{dx} = \frac{dy}{dt} \div \frac{dx}{dt}$ or $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$.
46. Derivative of one function $f(x)$ with respect to another function $g(x)$, take $u = f(x)$ and $v = g(x)$ and then find $\frac{du}{dv} = \frac{f'(x)}{g'(x)}$.
47. If $f(x, y) = c$ is a function then $\frac{dy}{dx} = \frac{-\partial f / \partial x}{\partial f / \partial y}$.
48. If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, then $\frac{dy}{dx} = \frac{-(ax + hy + g)}{(hx + by + f)}$.
49. If $x^m y^n = a^{m+n}$, then $\frac{dy}{dx} = \frac{-my}{nx}$.
50. If $f(x, y) = 0$ is a homogeneous function then $\frac{dy}{dx} = \frac{y}{x}$.
51. $f(x) = |x|$ is not differentiable at one point $x = 0$.
52. $f(x) = |x| + |x - 1|$ is not differentiable at two points $x = 0, 1$.
53. $f(x) = |x| + |x - 1| + |x - 2|$ is not differentiable at three points $x = 0, 1, 2$.
54. $x \sin \frac{1}{x}$, $x \cos \frac{1}{x}$ are continuous but not differentiable at $x = 0$.
55. If $f(x + y) = f(x) + f(y)$, then $f'(x) = f'(0)$.

• **Differentiation by Trigonometric Substitution:**

Sl No.	Function	Substitution	Conditions
1	$\sqrt{a^2 - x^2}$	$x = a \sin q$, or $x = a \cos q$	$q \in [0, \pi/2]$
2	$\sqrt{x^2 + a^2}$	$x = a \tan q$, or $x = a \cot q$	$q \in [0, \pi/2]$
3	$\sqrt{x^2 - a^2}$	$x = a \sec q$, or $x = a \csc q$	$q \in [0, \pi/2]$
4	$\sqrt{\frac{a-x}{a+x}}$	$x = a \cos 2q$	$q \in [0, \pi/2]$
5	$\sqrt{\frac{a^2 - x^2}{a^2 + x^2}}$	$x^2 = a^2 \cos 2q$	$q \in [0, \pi/2]$
6	$\sqrt{ax - x^2}$	$x = a \sin^2 q$	$q \in [0, \pi/2]$
7	$\sqrt{\frac{x}{a+x}}$	$x = a \tan^2 q$	$q \in [0, \pi/2]$
8	$\sqrt{\frac{x}{a-x}}$	$x = a \sin^2 q$	$q \in [0, \pi/2]$
9	$\sqrt{(x-a)(x-b)}$	$x = a \sec^2 q - b \tan^2 q$	$q \in [0, \pi/2]$
10	$\sqrt{(x-a)(b-x)}$	$x = a \cos^2 q + b \sin^2 q$	$q \in [0, \pi/2]$

• **Differentiation of Infinite Series:**

(a) If $y = \sqrt{f(x) + \sqrt{f(x) + \sqrt{f(x) + \dots \infty}}}$, then

$$\Rightarrow y = \sqrt{f(x) + y}.$$

$$\Rightarrow y^2 = f(x) + y.$$

$$\Rightarrow y^2 - y = f(x) \text{ on differentiating both side wrt } x \text{ we get}$$

$$\Rightarrow 2y \frac{dy}{dx} - \frac{dy}{dx} = f'(x).$$

$$\Rightarrow (2y - 1) \frac{dy}{dx} = f'(x).$$

$$\Rightarrow \frac{dy}{dx} = \frac{f'(x)}{2y - 1}.$$

(b) If $y = f(x)^{f(x)^{f(x)^{\dots\infty}}}$, then

$$\Rightarrow y = f(x)^y.$$

$\Rightarrow \log y = y \log f(x)$ on differentiating both side wrt x we get

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} = y \frac{f'(x)}{f(x)} + f(x) \frac{dy}{dx}.$$

$$\Rightarrow \frac{1}{y} \frac{dy}{dx} - f(x) \frac{dy}{dx} = y \frac{f'(x)}{f(x)}.$$

$$\Rightarrow \left(\frac{1}{y} - f(x) \right) \frac{dy}{dx} = y \frac{f'(x)}{f(x)}.$$

$$\Rightarrow \frac{1 - yf(x)}{y} \frac{dy}{dx} = y \frac{f'(x)}{f(x)}.$$

$$\Rightarrow \frac{dy}{dx} = \frac{y^2 f'(x)}{f(x)[1 - yf(x)]}.$$

(c) If $y = f(x) + \frac{1}{f(x) + \frac{1}{f(x) + \frac{1}{f(x) + \frac{1}{f(x) \dots \infty}}}}$.

$$\Rightarrow y = f(x) + \frac{1}{y}.$$

$\Rightarrow y^2 = yf(x) + 1$ on differentiating both side wrt x we get

$$\Rightarrow 2y \frac{dy}{dx} = yf'(x) + f(x) \frac{dy}{dx}.$$

$$\Rightarrow 2y \frac{dy}{dx} - f(x) \frac{dy}{dx} = yf'(x).$$

$$\Rightarrow (2y - f(x)) \frac{dy}{dx} = yf'(x).$$

$$\Rightarrow \frac{dy}{dx} = \frac{yf'(x)}{2y - f(x)}.$$

(d) $f''(a) = \lim_{h \rightarrow 0} \frac{f(a+2h) - 2f(a+h) + f(a)}{h^2}.$

(e) $f^{(n)}(a) = \lim_{h \rightarrow 0} \frac{f(a+nh) - {}^nC_1 f(a+(n-1)h) + {}^nC_2 f(a+(n-2)h) - \dots + (-1)^n f(a)}{h^n}.$

(f) $\left\{ \frac{d}{dx} f^{-1}(x) \right\}_{x=f(a)} = \frac{1}{\left\{ \frac{d}{dx} f(x) \right\}_{x=a}}.$

• **Logarithmic Differentiation:**

- (a) If $y = f_1(x) \times f_2(x) \times f_3(x) \times \dots$ or $y = \frac{f_1(x) \times f_2(x) \times \dots}{g_1(x) \times g_2(x) \times \dots}$ then first take logarithm both side and then differentiate.
- (b) If $y = u^v$ then $\frac{dy}{dx} = u^v \frac{d}{dx}(v \log u)$.
- (c) If $y = x^x$ then $\frac{dy}{dx} = x^x \frac{d}{dx}(x \log x) = x^x \left(x \times \frac{1}{x} + \log x \times 1 \right) = x^x (1 + \log x)$.
- (d) $\frac{d}{dx}(\log f(x)) = \frac{1}{f(x)} \frac{d}{dx}(f(x))$.
- (e) $\frac{d}{dx}(|u|) = \frac{u}{|u|} \frac{du}{dx}$.
- (f) $\frac{d}{dx}(a^{f(x)}) = a^{f(x)} \log a \times \frac{d}{dx}(f(x))$.

• **Leibnitz Theorem for nth derivative of product of two functions:**

$$\frac{d^n}{dx^n}(u.v) = u_n v + {}^nC_1 u_{n-1} v_1 + {}^nC_2 u_{n-2} v_2 + {}^nC_3 u_{n-3} v_3 + \dots + u v_n.$$

- (a) $\frac{d^n}{dx^n}(x^n) = n!$.
- (b) $\frac{d^n}{dx^n}\left(\frac{1}{x}\right) = \frac{(-1)^n n!}{x^{n+1}}$.
- (c) $\frac{d^n}{dx^n}(\log x) = \frac{(-1)^{n-1} (n-1)!}{x^n}$.
- (d) $\frac{d^n}{dx^n}(\sin x) = \sin\left(x + n \frac{\pi}{2}\right)$.
- (e) $\frac{d^n}{dx^n}(\cos x) = \cos\left(x + n \frac{\pi}{2}\right)$.
- (f) $\frac{d^n}{dx^n}(e^{mx}) = m^n e^{mx}$.
- (g) $\frac{d^n}{dx^n}(a^x) = a^x (\log a)^n$.
- (h) $\frac{d^n}{dx^n}((ax+b)^m) = \begin{cases} m(m-1)(m-2)\dots(m-n+1)(ax+b)^{m-n} a^n & n < m \\ m! a^m & n = m \\ 0 & n > m \end{cases}$.

$$\begin{aligned} \text{(i)} \quad & \frac{d^n}{dx^n} \left(\frac{1}{ax+b} \right) = \frac{(-1)^n a^n \cdot n!}{(ax+b)^{n+1}}. \\ \text{(j)} \quad & \frac{d^n}{dx^n} (\log(ax+b)) = \frac{(-1)^{n-1} a^n \cdot (n-1)!}{(ax+b)^n}. \\ \text{(k)} \quad & \frac{d^n}{dx^n} (\sin(ax+b)) = a^n \sin \left(ax+b + n \frac{\pi}{2} \right). \\ \text{(l)} \quad & \frac{d^n}{dx^n} (\cos(ax+b)) = a^n \cos \left(ax+b + n \frac{\pi}{2} \right). \end{aligned}$$

- **Partial Differentiation**: The partial differentiation of $f(x, y)$ with respect to x is the ordinary derivative of $f(x, y)$ with respect to x by taking y as constant and is denoted by $\frac{\partial f(x, y)}{\partial x} = D_x f(x, y) = f_x = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$. Similarly the partial differentiation of $f(x, y)$ with respect to y is the ordinary derivative of $f(x, y)$ with respect to y by taking x as constant and is denoted by $\frac{\partial f(x, y)}{\partial y} = D_y f(x, y) = f_y = \lim_{k \rightarrow 0} \frac{f(x, y+k) - f(x, y)}{k}$.

- **Higher Partial Derivatives**: Let $f(x, y)$ be a function of two variables such that $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}$ both exists.

$$\begin{aligned} \text{(a)} \quad & \text{The partial derivative of } \frac{\partial f}{\partial x} \text{ with respect to } x \text{ is denoted by } \frac{\partial^2 f}{\partial x^2} = f_{xx}. \\ \text{(b)} \quad & \text{The partial derivative of } \frac{\partial f}{\partial x} \text{ with respect to } y \text{ is denoted by } \frac{\partial^2 f}{\partial x \partial y} = f_{xy}. \\ \text{(c)} \quad & \text{The partial derivative of } \frac{\partial f}{\partial y} \text{ with respect to } x \text{ is denoted by } \frac{\partial^2 f}{\partial y \partial x} = f_{yx}. \\ \text{(d)} \quad & \text{The partial derivative of } \frac{\partial f}{\partial y} \text{ with respect to } y \text{ is denoted by } \frac{\partial^2 f}{\partial y^2} = f_{yy}. \\ \text{(e)} \quad & \frac{\partial^2 f}{\partial y \partial x} = f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = f_{xy}. \end{aligned}$$

- **Homogeneous function of degree n**: A function $f(x, y)$ is said to be a homogeneous function of degree n if $f(tx, ty) = t^n f(x, y)$.

- **Euler's Theorem on Homogeneous Function:** If $f(x, y)$ is a homogeneous function of degree n , then:

$$(a) \quad x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = nf .$$

$$(b) \quad x \frac{\partial^2 f}{\partial x^2} + y \frac{\partial^2 f}{\partial x \partial y} = (n-1) \frac{\partial f}{\partial x} .$$

$$(c) \quad x \frac{\partial^2 f}{\partial y \partial x} + y \frac{\partial^2 f}{\partial y^2} = (n-1) \frac{\partial f}{\partial y} .$$

$$(d) \quad x^2 \frac{\partial^2 f}{\partial x^2} + 2xy \frac{\partial^2 f}{\partial x \partial y} + y^2 \frac{\partial^2 f}{\partial y^2} = n(n-1) f .$$
